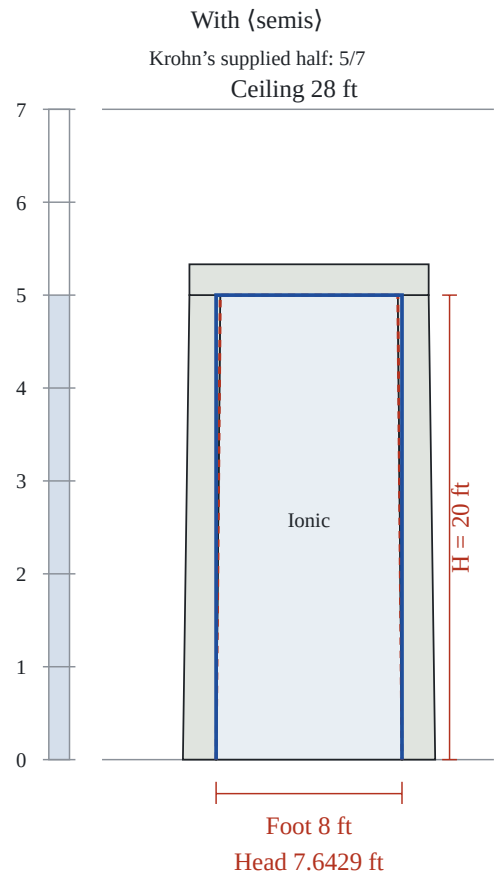
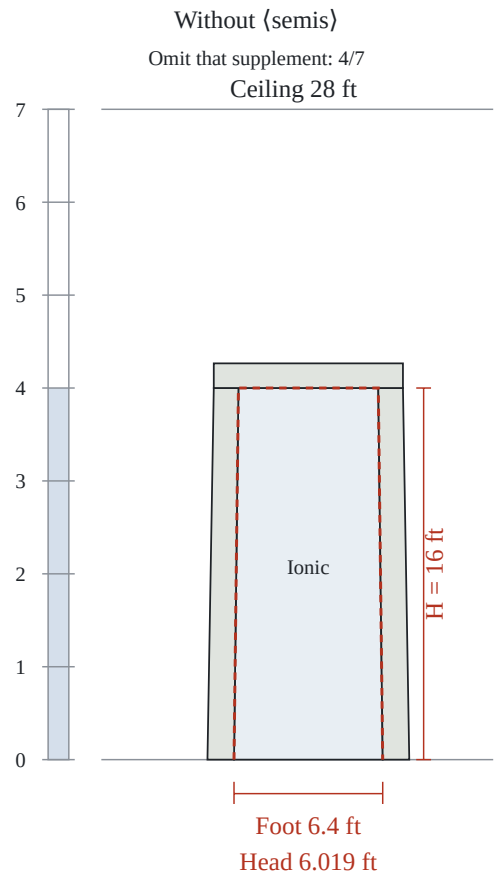
A half changes the doorway

Vitruvius's doorway proportions, read with and without the supplied half. Ceiling 28 Roman feet; Ionic; selected reading with ⟨semis⟩ (5/7). Selected opening height 20 ft. Over 16 through 25 ft: one quarter of a jamb.

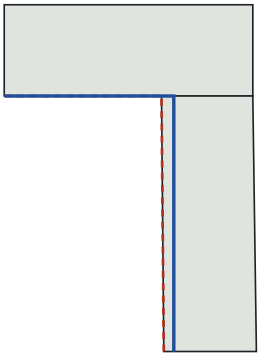


Selected reading. Solid blue: your trial. Dashed red: computed rule.



The other height reading, with its own contraction band.

Upper right corner, enlarged



Both axes enlarged equally. Stone follows the rule.
Trial differs by 0.1786 ft at this corner.

Measurements for the selected reading

Measure	Roman feet	Basis
Ceiling height	28	chosen
Opening height	20	5/7 of ceiling; supplied semis
Opening width at foot	8	2/5 of opening height
Jamb width at foot	1.4286	opening height / 14; IV.6.3
Jamb width at head	1.3265	13/14 of lower jamb width; IV.6.2
Whole opening contraction	0.3571	lower jamb width / 4; applied to total width
Opening width at head	7.6429	lower width minus whole contraction
Each upper corner inset	0.1786	half the whole contraction
Lintel height	1.3265	upper jamb width; IV.6.2
Trial upper width	8	learner trial; not an ancient prescription
Trial upper width minus rule	0.3571	positive means too wide

Choices and limits

- A reading of Vitruvius IV.6.1–3, not a reconstruction of a particular ancient doorway.
- The height comparison omits only Krohn's second, supplied semis. Other editorial choices stay fixed.
- The opening contraction is subtracted once from the whole width, half from each side; the lower jamb supplies the unit.
- The shared endpoints 16, 25 and 30 feet belong to the lower contraction band on this desk. The text does not uniquely settle these endpoints.
- The Ionic jamb is H/14, as printed on Krohn p. 89. The Doric jamb uses that same value as a stated drawing assumption across Krohn's lacuna.
- Jamb widths are measured horizontally on the face. The upper jamb retains 13/14 of the lower width. The lintel height equals the upper jamb width.
- The casing follows the computed rule. Solid blue is the learner's trial, which can cross into that casing. Both main drawings share a scale; the corner enlargement uses the same scale on both axes.
- Mouldings, frieze, consoles, door leaves, wall depth, optical perception and structural performance are not modelled. Roman feet are used without a modern metric conversion.

Words in the drawing

lvmen · The opening. luminis: ‘of the opening’. The clear space through the wall. The Doric width sentence measures this empty space at its foot; adding two jambs would measure something else. The more general Latin sense is ‘light’. (IV.6.1, p. 88)

antepagmentvm · The casing. antepagmenti: ‘of the casing’. The doorway surround; here the side pieces are the jambs. Its width supplies the unit for narrowing the opening. Its own narrowing follows a different fraction, one fourteenth. (IV.6.1–3, pp. 88–89)

svpercilivm · The lintel. supercilii: ‘of the lintel’. An ‘eyebrow’ above the doorway. The text relates its thickness to the upper end of the casing. The drawing shows that relation but leaves the mouldings and the frieze above it aside. (IV.6.2, p. 88)

lacvnaria · The coffered ceiling. ad lacunaria: ‘to the coffered ceiling’. This is the upper end of the height being divided. The opening height is a fraction of the distance from the pavement to this ceiling; it is not a fraction of the columns or the roof ridge. (IV.6.1, p. 88)

contrahatvr · Let it narrow. Present passive subjunctive of contraho. A prescription: ‘let it be drawn together’. Identify what is being narrowed and what supplies the fraction. The opening narrows by a fraction of a jamb; the jamb narrows by a fraction of itself. (IV.6.1–2, p. 88)

valvae · The door leaves. valvarum: ‘of the doors’. The doors that close the opening. The Latin distinguishes these from the lumen. Door-leaf construction appears later in IV.6; the drawing deliberately leaves the opening empty. (IV.6.1, p. 88)

θύρωμα · A Greek word inside the Latin. Greek θυρωμάτων, genitive plural of θύρωμα. Vitruvius names Doric, Ionic and Attic kinds of doorway with this Greek genitive plural, ‘of doorways’. Krohn prints the word in Latin letters on p. 88; the accented Greek here is the dictionary form, not a quotation of Greek lettering on the scan. (IV.6.1, p. 88)

Read the instructions

IV.6.1 · Three doorway kinds

genera svnt enim thyromaton haec: doricvm, ionicvm, atticvrges.

For these are the kinds of doorway: Doric, Ionic and Attic.

Thyromaton is Greek in Latin letters. This desk draws only the Doric and Ionic proportions; it does not supply an Attic doorway.

Krohn 1912, printed p. 88.

IV.6.1 · A supplied half

lvmen avtem thyretri constitvatvr sic, vti qvae altitvdo aedis a pavimento ad lacvnaria fverit, dividatvr in partes tres semis et ex eis dvae partes ⟨semis⟩ lvmini valvarvm altitvdine constitvantvr.

Set the opening of the doorway as follows: divide the temple's height from the pavement to the coffered ceiling into three and a half parts; of these, let two ⟨and a half⟩ parts establish the height of the opening for the doors.

Krohn prints the second semis inside angle brackets. His apparatus credits its addition to Rode. With it, (2½)/(3½) = 5/7; omit only that supplement and 2/(3½) = 4/7. These are two readings of this instruction, not two complete recovered buildings or two uniform manuscript traditions.

Krohn 1912, printed p. 88.

IV.6.1 · The lower opening

haec avtem dividatvr in partes xii et ex eis qvinqve et dimidia latitvdo lvminis fiat in imo.

Divide this [height] into twelve parts, and let five and a half of them be the width of the opening at the bottom.

The Doric lower opening is 5½/12 = 11/24 of its height. Luminis means 'of the opening': this width excludes both stone jambs.

Krohn 1912, printed p. 88.

IV.6.1 · The opening's contraction

et in svmmo contrahatvr, si erit lvmen ab imo ad sedecim pedes, antepagmenti iii parte; ⟨si⟩ xvi pedvm ad xxv, svperior pars lvminis contrahatvr antepagmenti parte iiii; si ab pedibvs xxv ad xxx, svmma pars contrahatvr antepagmenti parte viii. reliqva, qvo altiora ervnt, ad perpendicvlvm videntvr oportere conlocari.

At the top let it narrow, if the opening is up to sixteen feet high from the bottom, by a third of the casing; from sixteen feet to twenty-five, let the upper part of the opening narrow by a quarter of the casing; from twenty-five to thirty feet, by an eighth. The remaining, taller ones seem to require perpendicular sides.

The fraction is of the jamb, not the opening or its height. This desk subtracts it once from the whole opening width and splits that loss equally between the two sides. The ranges share endpoints: this desk keeps 16, 25 and 30 in the lower band. That boundary convention and the symmetric allocation are modelling decisions. No interpolation between bands is implied.

Krohn 1912, printed p. 88.

IV.6.2 · The casing's own taper

ipsa avtem antepagmenta * * * * contrahantvr in svmmo svae crassitvdinis xiiii parte. svpercilii crassitvdo, qvanta antepagmentorvm in svmma parte erit crassitvdo.

The casings themselves [gap in the text] are to narrow at the top by a fourteenth of their own width. The lintel's thickness is to be as much as the casings' width at their upper end.

The stars reproduce Krohn's marked lacuna, not a passage omitted for brevity. The jamb loses 1/14 of its own lower width, independently of the opening's contraction. The desk draws the lintel height as that upper jamb width and omits the overdoor mouldings.

Krohn 1912, printed p. 88.

IV.6.3 · The Ionic opening and casing

Sin avtem ionico genere fvtvra ervnt, lvmen altvm ad evndem modvm qvemadmodvm in doricis fieri videtvr. latitvdo constitvatvr, vt altitvdo dividatvr in partes dvas et dimidiam, eivsqve partis vnivs ima lvminis fiat latitvdo. contractvrae ita vti in doricis. crassitvdo antepagmentorvm ⟨ex⟩ altitvdine lvminis in fronte xiiii parte

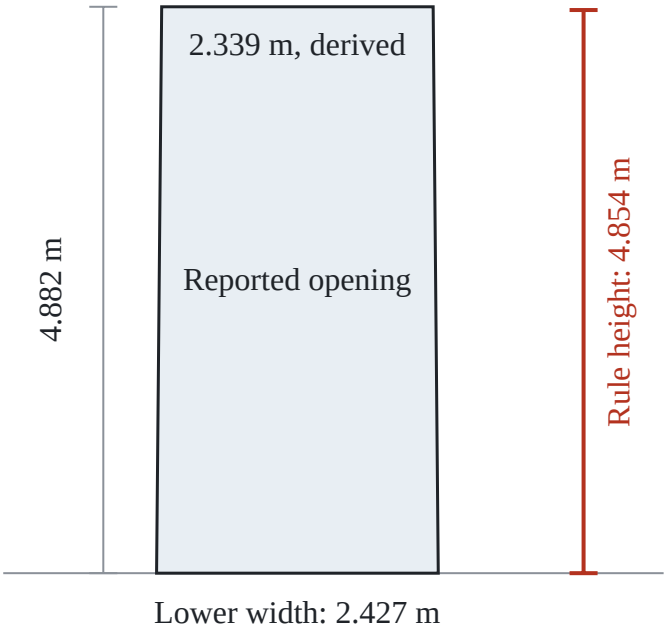
If they are to be Ionic, the opening's height seems to be made in the same way as in Doric. Set the width by dividing the height into two and a half parts and giving the lower opening one of them. The contractions are as in Doric. The width of the casings on their face is a fourteenth of the opening's height.

The print on p. 89 reads XIIII, fourteen, rather than XII, twelve. This fixes the Ionic lower jamb at H/14. For Doric, the desk explicitly borrows that Ionic value across the lacuna; it does not silently supply missing Latin. Ionic lower width is H/2½ = 2H/5. Consoles, fasciae and door leaves are outside this diagram.

Krohn 1912, printed p. 89.

Almost a double square

The North Door of the Erechtheion, as reported by the survey published in *The Erechtheum* (1927), pp. 98–99. This comparison keeps the survey's original-opening interpretation and its omitted later linings visible.



Metres from the survey; no Roman foot conversion.

Exact double square, lower width matched

The other dimension is **28 mm shorter** than the reported opening.

At 2.427 m wide below, the rule gives 4.854 m high; the report gives 4.882 m.

Reported and calculated values · metres

Measure	Metres	Basis
Reported opening height	4.882	Metres; 1927 survey p. 99; later lining omitted

Measure	Metres	Basis
Reported lower opening width	2.427	Metres; 1927 survey p. 99
Reported inward movement of each jamb	0.044	Metres; rounded value on p. 99
Derived upper opening width	2.339	Reported lower width minus twice the reported jamb movement
Rule height under this alignment	4.854	Exact double square; width matched
Rule lower width under this alignment	2.427	1/2 of height
Rule height minus reported height	-0.028	Metres; positive means taller
Rule lower width minus reported lower width	0	Metres; positive means wider

Reported lower width ÷ height = 0.4971. Selected rule = 1/2 = 0.5.

What the comparison assumes

- The 1927 volume reports what it identifies as the original opening. Its pp. 98–99 explicitly leave later inner linings aside. These are not measurements of today's clear opening.
- The main reported dimensions are 4.882 m high and 2.427 m wide at the bottom; the inward movement of each jamb is reported as 0.044 m on p. 99. The derived upper width is 2.427 − 2 × 0.044 = 2.339 m.
- Straight lines join the reported endpoints in the drawing. No moulding profile, casing width, leaf, lintel thickness or wall depth is inferred.
- Only the dimensionless lower-width relation is compared. Matching the reported height leaves a lower-width difference; matching the lower width leaves a height difference. Neither the lost ceiling height nor the opening's age follows from this fit.
- The Vitruvian upper-width contraction bands are not applied to these metric data. The survey and ancient text have different units; no length for a Roman foot is assumed.
- The exact double square is a comparison to the survey's word 'essentially'. Printed dimensions are retained rather than adjusted to force equality. Extra decimal places in calculated outputs do not improve the survey's precision.

Keep working with the drawing

This file includes both experiments, the corner, measurements in separate Roman-foot and metric tables, Latin, new English translations, four complete source pages, four font subsets and licenses. It uses no scripts and opens offline. External links need a connection.

Reopen these choices in Embater.

Vitruvius, *De architectura*, ed. F. Krohn (Teubner, 1912), complete printed p. 88, PDF page 106. Public domain. Original scan. SHA-256 bd7e110dbd0db15f159cd48cde18910f9da0fb83128d58ae5beac673cbc115dd.

VI Ostiorum autem et eorum antepagmentorum in aedibus hae sunt rationes, uti primum constituatur, quot genera sint [futurae]. genera sunt enim thyromaton haec: doricum, ionicum, atticurges.

Horum ⟨dorici generis⟩ symmetriae conspiciuntur his rationibus, uti corona summa, quae supra antepagmentum superius inponitur, aeque librata sit capitulis summis columnarum, quae in pronao fuerint. lumen autem thyretri constituatur sic, uti quae altitudo aedis a pavimento ad lacunaria fuerit, dividatur in partes tres semis et ex eis duae partes ⟨semis⟩ lumini valvarum altitudine constituantur. haec autem dividatur in partes XII et ex eis 97 quinque et dimidia latitudo luminis fiat in imo. et in summo contrahatur, si erit lumen ab imo ad sedecim pedes, antepagmenti III parte; ⟨si⟩ XVI pedum ad XXV, superior pars luminis contrahatur antepagmenti parte IIII; si ab pedibus XXV ad XXX, summa pars contrahatur antepagmenti parte VIII. reliqua, quo altiora erunt, ad 2 perpendiculum videntur oportere conlocari. ipsa autem antepagmenta * * * * contrahantur in summo suae crassitudinis XIIII parte. supercilii crassitudo, quanta antepagmentorum in summa parte erit crassitudo. cymatium faciundum est antepagmenti parte sexta; proiectura autem, quanta est eius crassitudo. sculpendum est cymatium lesbium cum astragalo. supra cymatium, quod erit in supercilio, conlocandum est hyperthyrum crassitudine supercilii, et in eo scalpendum est cymatium doricum, astragalum lesbium sima scalptura. corona plana cum cymatio; proiectura autem eius erit quanta altitudo. super-

---

2 constituatur, quot genera sint *Kr*: constituantur quod (quo *GS*) genere sint futurae *x* (*de delendo* futurae *cf. pag. 253,26 et 266,2*) 4 attigurges *x* 5 *add. Joc* 7 inponetur *HS* 8 thyretri *Ro*: hypetri *x* 10 III semis *G* 11 partis *H add. Rode* lumini *Joc*: lumine *x* 13 latitudo H^cSG^c: altitudo *a. c. HG* 15 pedis *H* si *add. Schn* 16 quarta parte *G* 17 ad pedibus *H et a. ras. S*, a pedibus GS^c 23 faciendum *GS* 28 astragulū lespiū *S*

Vitruvius, *De architectura*, ed. F. Krohn (Teubner, 1912), complete printed p. 89, PDF page 107. Public domain. Original scan. SHA-256 bf953dd783c62d9ec44bfea2863697400d0f5b5219c961d8ed5b5da416d76a0b.

cilii, quod supra antepagmenta inponitur, dextra atque sinistra proiecturae sic sunt faciundae, uti crepidines ex-
(20) currant et in unguem ipso cymatio coniungantur.

Sin autem ionico genere futura erunt, lumen altum ad 3
5 eundem modum quemadmodum in doricis fieri videtur. latitudo constituatur, ut altitudo dividatur in partes duas et dimidiam, eiusque partis unius ima luminis fiat latitudo.
(25) contracturae ita uti in doricis. crassitudo antepagmentorum ⟨ex⟩ altitudine luminis in fronte XIIII parte, cy-
98 matium huius crassitudinis sexta. reliqua pars praeter cy-
11 matium dividatur in partes XII. harum trium prima corsa fiat cum astragalo, secunda quattuor, tertia quinque, et eae aeque corsae cum astragalis circumcurrant. hyper- 4
(5) thyra autem ad eundem modum componantur quemad-
15 modum in doricis. pro crepidinibus ancones, sive parotides vocantur, excalptae dextra ac sinistra praependeant ad imi supercilii libramentum, praeter folium. eae habeant in fronte crassitudinem ex antepagmenti tribus partibus,
(10) imo quarta parte graciliore quam superiora.

Fores ita compingantur, uti scapi cardinales sint ex
20 latitudine luminis totius XII parte. inter duos scapos tympana ex XII partibus habeant ternas partes. inpagibus 5 distributiones ita fient, uti divisis altitudinibus in partes V
(15) duae superiori, tres inferiori designentur. super medium
25 medii inpages conlocentur, ex reliquis alii in summo, alii in imo compingantur. altitudo inpagis fiat tympani tertia parte, cymatium sexta parte inpagis. scaporum latitudines inpagis dimidia parte, item replum de inpage dimidia et

1 ac sin. *G* 2 faciendae *G* 3 unguem *Schn*: ungue *x* 7 ima luminis *G et in ras. S*: simalum in his *H* 8 ctraiecturę *S* 9 ex *add. Kr* (*cf. 98, 8 et 11*) 11 dividatur *ed*: -ditur *x* harum *Joc*: horum *x* 12 et eae *Ro*: ex ea *x* 13 hyperthyra *Joc*: hypetra *HS*, hipetre *G* (-ę *G*c) 15 pro crepidinibus *Kr*: protis pedibus *x* parotides *Schn*: protides *HG*, ptides *S* 16 excalptae *Ro*: excalpta *H*, exscalpta *S*, etscalpta *G* 18 antepagmenti *Joc*: -tis *x* 19 imo *Kr*: inimo (*cum* in *add.*) *x* 21 altitudine *HS* 22 duodecī *S* inpagibus *ed*: inpaginibus *x* 24 III *G* 25 ex *ed*: et (&) *x* 27 scaphorum *HS*

The Erechtheum, ed. J. M. Paton; survey G. P. Stevens (Harvard University Press, 1927), complete printed p. 98, PDF page 128. Public domain. Original scan. SHA-256 c3927e320fdc5005877d76fbd140c6a0c1a7c2a9e84bfa1db25241892cfdd761.

IX. The North Door

[Plates XXV, XXXV, 3]

It has long been recognized [1] that the present inner jambs and lintel lining of the North Door are relatively late additions, not earlier, in fact, than the transformation of the temple into a church; they will, therefore, be passed over for the present, and the description will begin with an examination of the original jambs and lintel.[2]

JAMBS Both the jambs were held by T-dowels [3] in their lower ends, the lead being run in from the sides toward the wall (Plate XXV, 6, 7), — an operation which was possible only if the jambs were erected before the orthostates on the right and left of the door were in place. The boss on the north face of the east jamb near the top, which was not worked off in the

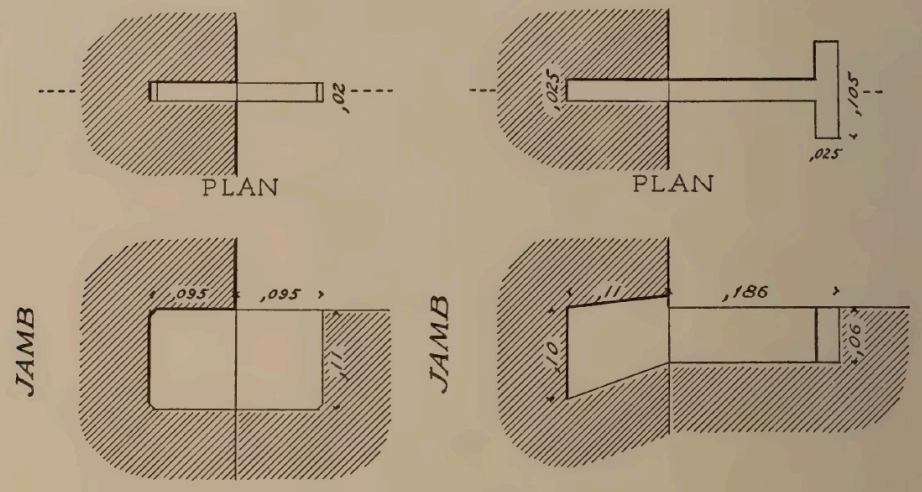


Figure 64. North Door: special forms of clamp (right) and of dowel (left) used to secure jambs

final dressing, would have been useful as an aid in steadying the jamb during the construction, although it may have been used merely in erecting the stone. From the top of the orthostates horizontal dowels, two on each side, of the form shown in Figure 64 (left), extended into the jambs and were sealed with lead. In courses 13, 15, and 17 two clamps,

[1] Kinnard (Stuart and Revett,[2] II, pp. 72, 73, note c) seems first to have called attention in print to the fact that the insertion of the later lining was caused by the breaking of the great lintel. Donaldson, *Ancient Doorways* (1818), text to pl. XXIII, first noted in print that these linings were late. See also Inwood, *ed. maj.*, p. 112. In order to bring out the effect of the door as originally designed the late linings have been omitted in Plate XXV. They are shown in Plates VII, X, and XI.

[2] Cf. on the North Door, R. W. Schultz, *J. H. S.*, XII, 1891, pp. 1–13, pls. I–III, and the correction of one important point by S. H. Barnsley, *ibid.*, pp. 381–383.

[3] The cuttings for these dowels are visible from the interior of the building, where the jambs and the sill have been hacked away in the search for metal.

The Erechtheum, ed. J. M. Paton; survey G. P. Stevens (Harvard University Press, 1927), complete printed p. 99, PDF page 129. Public domain. Original scan. SHA-256 c1cdb9daaa68cfd3f877c031adc34230015b9acaeac76c89431aeba2a60fa5ae.

modified from the usual ⊢ form (Fig. 64, right), extended from the top of each course into the back of the jamb and were sealed in the same way (PLATE XXV, 6). Similarly, in course 16 certainly, and probably in courses 14 (on the east side of the door), 12, and 11, one such clamp was fixed in the jamb from the axis of the top of the course. The upper ends of the jambs were fastened to the adjacent course (10) of the wall by two ⊢ clamps (0.31 m. long) on each side (Fig. 40). As in every course the clamp could be inserted and sealed with lead only before the next course was laid, the conclusion is inevitable: the jambs are contemporary with the erection of the building.[1] It may be added that the jambs are rabbeted to overlap the wall (PLATES XXV, 6, XXX, 18) and that the weathering and final dressing of the wall indicate that jambs with an overlap of this width (0.0207) m. have always existed. The reveals of the jambs are carefully finished for their entire width, which implies that there was no inner sheathing. The inclination of each jamb toward the axis of the door is 0.044 m. The original Greek door was 4.882 m. in height, and 2.427 m. in width at the bottom, so that its opening formed essentially a double square.

LINTEL

The lintel is now composed of two stones, one above the other, and together equal in height to two courses (8 and 9) of the wall (PLATES XXV, XXXV, 3; Fig. 98). Its lower surface is smooth, showing that there was no casing. The ends of the lintel are rough and without anathyrosis. The lower stone is fastened at the top of its east end into the block of the console by a dowel, similar to those used between orthostates and jambs, but of different dimensions; how its west end is secured cannot be seen. The upper stone was held in place by iron wedges between it and the blocks of the wall above. One of the latter has a dowel cutting at its end, which has no corresponding cutting in the lintel. West of the lintel is a block of irregular shape, its east end dressed roughly without anathyrosis.[2] On its outer face are cuttings for the two dowels [3] which held the console that was under the west end of the upper part of the lintel (Fig. 65).[4] East of the lintel a space of similar shape is filled by

[1] As urged by Barnsley, *l. c.*

[2] PLATE VII, and, for the stone as seen from the inside, Figure 40; cf. *J. H. S.*, XII, 1891, p. 2, fig. 1.

[3] These cuttings are only about 0.05 m. wide, and 0.09 m. high by 0.10 m. deep, so that heavy dowels of metal seem more probable than tenons of marble projecting from the console.

[4] It is not certain when the west console disappeared. In the British Museum (MSS., *Add.* 22,153, fol. 153) is a drawing made in 1818 by R. H. Sharpe of York, showing apparently the west corner of the door (Fig. 65). The console differs in details from the surviving one, but since there are errors in the rendering of the other mouldings, these differences may be due to the inaccuracy of the artist in preparing the finished drawing from his notes. This drawing is on tracing paper, and it is possible that it has been traced from a reversed drawing on heavy paper of the east corner, for it is not an uncommon architectural process to draw an object roughly, and then to make a clean and accurate tracing, which is turned upside down upon heavy paper and rubbed, thus making a clean reversed drawing upon the heavy paper. This drawing cannot, therefore, be cited as conclusive evidence for the existence of the console in 1818. On the other hand there is no direct statement that it had already been broken at that time. The portico then contained the Turkish powder magazine, which was protected by a concrete vault rising nearly to the top of the door and concealing in 1811 all but the upper part of the lintel (see Cockerell's drawing, *J. H. S.*, XXIX, 1909, pl. VII = Fig. 223). In 1818 things were somewhat better, for then Inwood (*ed. min.*, p. 15, to pl. XX) cleared away "part of the ground . . . over the bottom of the console and ornaments for the purpose of making casts of them." This was the east console, but Inwood elsewhere, Wolfe (see App. A, XVII), and Donaldson (*Ancient Doorways*, pl. XXIII) regularly refer to *consoles* in the plural, without a hint that one was missing; in fact Donaldson writes of the ceiling of the vault as "not reaching higher than the consoles, which were quite clear." After the Greek Revolution and the

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